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TECHNICAL SPECIFICATION

**Digital cellular telecommunications system (Phase 2+) (GSM);
Universal Mobile Telecommunications System (UMTS);
LTE;
Technical Specifications and Technical Reports relating to the Common IP Multimedia Subsystem (IMS)
(3GPP TS 21.202 version 16.1.0 Release 16)**



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ETSI

650 Route des Lucioles
F-06921 Sophia Antipolis Cedex - FRANCE

Tel.: +33 4 92 94 42 00 Fax: +33 4 93 65 47 16

Siret N° 348 623 562 00017 - APE 7112B
Association à but non lucratif enregistrée à la
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Foreword

This Technical Specification has been produced by the 3rd Generation Partnership Project (3GPP).

The contents of the present document are subject to continuing work within the TSG and may change following formal TSG approval. Should the TSG modify the contents of the present document, it will be re-released by the TSG with an identifying change of release date and an increase in version number as follows:

Version x.y.z

where:

- x the first digit:
  - 1 presented to TSG for information;
  - 2 presented to TSG for approval;
  - 3 or greater indicates TSG approved document under change control.
- y the second digit is incremented for all changes of substance, i.e. technical enhancements, corrections, updates, etc.
- z the third digit is incremented when editorial only changes have been incorporated in the document.

1 Scope

The present document identifies the 3GPP Technical Specifications and Technical Reports specifically relating to the Common IP Multimedia Subsystem (IMS) maintained by 3GPP. Standards organizations adopting the Common IP Multimedia Subsystem (IMS) might not need to use all listed specifications.

2 References

The following documents contain provisions which, through reference in this text, constitute provisions of the present document.

- References are either specific (identified by date of publication, edition number, version number, etc.) or non-specific.
- For a specific reference, subsequent revisions do not apply.
- For a non-specific reference, the latest version applies. In the case of a reference to a 3GPP document (including a GSM document), a non-specific reference implicitly refers to the latest version of that document *in the same Release as the present document.*

[1] 3GPP TR 21.905: "Vocabulary for 3GPP Specifications".

[2] 3GPP TR 21.900: "Technical specification group working methods".

3 Definitions, symbols and abbreviations

3.1 Definitions

For the purposes of the present document, the terms and definitions given in TR 21.905 [1] apply.

3.2 Symbols

(None)

3.3 Abbreviations

For the purposes of the present document, the abbreviations given in TR 21.905 [1] and the following apply. An abbreviation defined in the present document takes precedence over the definition of the same abbreviation, if any, in TR 21.905 [1].

NGN	Next Generation Network
R1	Release 1
R2	Release 2
TISPAN	Telecommunications and Internet converged Services and Protocols for Advanced Networks

4 General

The numbering scheme for specifications is described in 3GPP TR 21.900 [2].

5 Specifications and Reports

NOTE 1: The "for publication?" column of the table below indicates whether or not the documents are intended for adoption by the partner Standards Development Organizations as their own publications. Those marked "no" are internal working documents of the 3GPP TSGs.

NOTE 2: Some of the algorithm specifications in the 35.-series are available only under licence.

NOTE 3: "Type" indicates Technical Specification (TS) or Technical Report (TR).

NOTE 4: For definition of "freezing" of specifications (last two columns), see 3GPP TR 21.900 [2].

The table below contains all Common IMS specs pertaining to Release 16.

Type	Number	Title	Group	For publication?
TS	21.111	USIM and IC card requirements	C6	Yes
TR	21.905	Vocabulary for 3GPP Specifications	SP	Yes
TS	22.030	Man-Machine Interface (MMI) of the User Equipment (UE)	S1	Yes
TS	22.041	Operator Determined Barring (ODB)	S1	Yes
TS	22.071	Location Services (LCS); Service description; Stage 1	S1	Yes
TS	22.101	Service aspects; Service principles	S1	Yes
TS	22.105	Services and service capabilities	S1	Yes
TS	22.115	Service aspects; Charging and billing	S1	Yes
TS	22.140	Multimedia Messaging Service (MMS); Stage 1	S1	Yes
TS	22.141	Presence service; Stage 1	S1	Yes
TS	22.153	Multimedia priority service	S1	Yes
TS	22.173	IP Multimedia Core Network Subsystem (IMS) Multimedia Telephony Service and supplementary services; Stage 1	S1	Yes
TS	22.174	Push Service; Service aspects; Stage 1	S1	Yes
TS	22.182	Customized Alerting Tones (CAT) requirements; Stage 1	S1	Yes
TS	22.183	Customized Ringing Signal (CRS) requirements; Stage 1	S1	Yes
TS	22.228	Service requirements for the Internet Protocol (IP) multimedia core network subsystem (IMS); Stage 1	S1	Yes
TS	22.250	IP Multimedia Subsystem (IMS) Group Management; Stage 1	S1	Yes
TS	22.279	Combined Circuit Switched (CS) and IP Multimedia Subsystem (IMS) sessions; Stage 1	S1	Yes
TS	22.340	IP Multimedia Subsystem (IMS) messaging; Stage 1	S1	Yes
TR	22.979	Feasibility study on combined Circuit Switched (CS) calls and IP Multimedia Subsystem (IMS) sessions	S1	Yes
TS	23.141	Presence service; Architecture and functional description	S2	Yes
TS	23.167	IP Multimedia Subsystem (IMS) emergency sessions	S2	Yes
TS	23.204	Support of Short Message Service (SMS) over generic 3GPP Internet Protocol (IP) access; Stage 2	S2	Yes
TS	23.218	IP Multimedia (IM) session handling; IM call model; Stage 2	C1	Yes
TS	23.228	IP Multimedia Subsystem (IMS); Stage 2	S2	Yes
TS	23.333	Multimedia Resource Function Controller (MRFC) - Multimedia Resource Function Processor (MRFP) Mp interface: Procedures descriptions	C4	Yes
TS	23.334	IP Multimedia Subsystem (IMS) Application Level Gateway (IMS-ALG) - IMS Access Gateway (IMS-AGW) interface: Procedures descriptions	C4	Yes
TR	23.794	Study on enhanced IP Multimedia Subsystem (IMS) to 5GC integration	S2	Yes
TS	24.141	Presence service using the IP Multimedia (IM) Core Network (CN) subsystem; Stage 3	C1	Yes
TS	24.147	Conferencing using the IP Multimedia (IM) Core Network (CN) subsystem; Stage 3	C1	Yes
TS	24.173	IMS Multimedia telephony communication service and supplementary services; Stage 3	C1	Yes
TS	24.174	Support of multi-device and multi-identity in the IP Multimedia Subsystem (IMS); Stage 3	C1	Yes
TS	24.175	Management Object (MO) for multi-device and multi-identity in the IP Multimedia Subsystem (IMS)	C1	Yes
TS	24.182	IP Multimedia Subsystem (IMS) Customized Alerting Tones (CAT); Protocol specification	C1	Yes
TS	24.229	IP multimedia call control protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP); Stage 3	C1	Yes
TS	24.238	Session Initiation Protocol (SIP) based user configuration; Stage 3	C1	Yes
TS	24.239	Flexible Alerting (FA) using IP Multimedia (IM) Core Network (CN) subsystem; Protocol specification	C1	Yes
TS	24.247	Messaging service using the IP Multimedia (IM) Core Network (CN) subsystem; Stage 3	C1	Yes

Type	Number	Title	Group	For publication?
TS	24.315	IP Multimedia Subsystem (IMS) Operator Determined Barring (ODB); Stage 3: protocol specification	C1	Yes
TS	24.341	Support of SMS over IP networks; Stage 3	C1	Yes
TS	24.604	Communication Diversion (CDIV) using IP Multimedia (IM) Core Network (CN) subsystem; Protocol specification	C1	Yes
TS	24.605	Conference (CONF) using IP Multimedia (IM) Core Network (CN) subsystem; Protocol specification	C1	Yes
TS	24.606	Message Waiting Indication (MWI) using IP Multimedia (IM) Core Network (CN) subsystem; Protocol specification	C1	Yes
TS	24.607	Originating Identification Presentation (OIP) and Originating Identification Restriction (OIR) using IP Multimedia (IM) Core Network (CN) subsystem; Protocol specification	C1	Yes
TS	24.608	Terminating Identification Presentation (TIP) and Terminating Identification Restriction (TIR) using IP Multimedia (IM) Core Network (CN) subsystem; Protocol specification	C1	Yes
TS	24.610	Communication HOLD (HOLD) using IP Multimedia (IM) Core Network (CN) subsystem; Protocol specification	C1	Yes
TS	24.611	Anonymous Communication Rejection (ACR) and Communication Barring (CB) using IP Multimedia (IM) Core Network (CN) subsystem; Protocol specification	C1	Yes
TS	24.615	Communication Waiting (CW) using IP Multimedia (IM) Core Network (CN) subsystem; Protocol Specification	C1	Yes
TS	24.616	Malicious Communication Identification (MCID) using IP Multimedia (IM) Core Network (CN) subsystem; Protocol specification	C1	Yes
TS	24.628	Common Basic Communication procedures using IP Multimedia (IM) Core Network (CN) subsystem; Protocol specification	C1	Yes
TS	24.629	Explicit Communication Transfer (ECT) using IP Multimedia (IM) Core Network (CN) subsystem; Protocol specification	C1	Yes
TS	24.642	Completion of Communications to Busy Subscriber (CCBS) and Completion of Communications by No Reply (CCNR) using IP Multimedia (IM) Core Network (CN) subsystem; Protocol specification	C1	Yes
TS	24.647	Advice Of Charge (AOC) using IP Multimedia (IM) Core Network (CN) subsystem	C1	Yes
TS	24.654	Closed User Group (CUG) using IP Multimedia (IM) Core Network (CN) subsystem, Protocol Specification	C1	Yes
TR	24.930	Signalling flows for the session setup in the IP Multimedia core network Subsystem (IMS) based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP); Stage 3	C1	Yes
TR	26.954	Test plan for speech quality and delay through a headset electrical interface	S4	Yes
TS	29.162	Interworking between the IM CN subsystem and IP networks	C3	Yes
TS	29.163	Interworking between the IP Multimedia (IM) Core Network (CN) subsystem and Circuit Switched (CS) networks	C3	Yes
TS	29.165	Inter-IMS Network to Network Interface (NNI)	C3	Yes
TS	29.212	Policy and Charging Control (PCC); Reference points	C3	Yes
TS	29.213	Policy and charging control signalling flows and Quality of Service (QoS) parameter mapping	C3	Yes
TS	29.214	Policy and charging control over Rx reference point	C3	Yes
TS	29.215	Policy and Charging Control (PCC) over S9 reference point; Stage 3	C3	Yes
TS	29.228	IP Multimedia (IM) Subsystem Cx and Dx Interfaces; Signalling flows and message contents	C4	Yes
TS	29.229	Cx and Dx interfaces based on the Diameter protocol; Protocol details	C4	Yes
TS	29.232	Media Gateway Controller (MGC) - Media Gateway (MGW) interface; Stage 3	C4	Yes
TS	29.238	Interconnection Border Control Functions (IBCF) - Transition Gateway (TrGW) interface, Ix interface; Stage 3	C4	Yes
TS	29.292	Interworking between the IP Multimedia (IM) Core Network (CN) subsystem (IMS) and MSC Server for IMS Centralized Services (ICS)	C3	Yes
TS	29.311	Service Level Interworking (SLI) for messaging services	C3	Yes
TS	29.328	IP Multimedia (IM) Subsystem Sh interface; Signalling flows and message contents	C4	Yes

Type	Number	Title	Group	For publication?
TS	29.329	Sh interface based on the Diameter protocol; Protocol details	C4	Yes
TS	29.333	Multimedia Resource Function Controller (MRFC) - Multimedia Resource Function Processor (MRFP) Mp interface; Stage 3	C4	Yes
TS	29.334	IMS Application Level Gateway (IMS-ALG) – IMS Access Gateway (IMS-AGW); Iq Interface; Stage 3	C4	Yes
TS	29.541	5G System; Network Exposure (NE) function services for Non-IP Data Delivery (NIDD) and Short Message Services (SMS); Stage 3	C4	Yes
TS	29.658	SIP Transfer of IP Multimedia Service Tariff Information; Protocol specification	C3	Yes
TS	31.101	UICC-terminal interface; Physical and logical characteristics	C6	Yes
TS	31.103	Characteristics of the IP Multimedia Services Identity Module (ISIM) application	C6	Yes
TS	31.115	Secured packet structure for (Universal) Subscriber Identity Module (U)SIM Toolkit applications	C6	Yes
TS	31.116	Remote APDU Structure for (U)SIM Toolkit applications	C6	Yes
TS	31.133	IP Multimedia Services Identity Module (ISIM) Application Programming Interface (API); ISIM API for Java Card™	C6	Yes
TS	32.240	Telecommunication management; Charging management; Charging architecture and principles	S5	Yes
TS	32.260	Telecommunication management;Charging management;IP Multimedia Subsystem (IMS) charging	S5	Yes
TS	32.275	Telecommunication management; Charging management; MultiMedia Telephony (MMTel) charging	S5	Yes
TS	32.276	Telecommunication management; Charging management; Voice Call Service (VCS) charging	S5	Yes
TS	32.281	Telecommunication management; Charging management; Announcement service	S5	Yes
TS	32.290	Telecommunication management; Charging management; 5G system; Services, operations and procedures of charging using Service Based Interface (SBI)	S5	Yes
TS	32.291	Telecommunication management; Charging management; 5G system, charging service; Stage 3	S5	Yes
TS	32.295	Telecommunication management; Charging management; Charging Data Record (CDR) transfer	S5	Yes
TS	32.297	Telecommunication management; Charging management; Charging Data Record (CDR) file format and transfer	S5	Yes
TS	32.298	Telecommunication management; Charging management; Charging Data Record (CDR) parameter description	S5	Yes
TS	32.299	Telecommunication management; Charging management; Diameter charging applications	S5	Yes
TS	33.141	Presence service; Security	S3	Yes
TS	33.203	3G security; Access security for IP-based services	S3	Yes
TS	33.210	Network Domain Security (NDS); IP network layer security	S3	Yes
TS	33.328	IP Multimedia Subsystem (IMS) media plane security	S3	Yes
TS	34.229-1	Internet Protocol (IP) multimedia call control protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP); User Equipment (UE) conformance specification; Part 1: Protocol conformance specification	R5	Yes
TS	34.229-2	Internet Protocol (IP) multimedia call control protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP); User Equipment (UE) conformance specification; Part 2: Implementation Conformance Statement (ICS) specification	R5	Yes
TS	34.229-3	Internet Protocol (IP) multimedia call control protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP); User Equipment (UE) conformance specification; Part 3: Abstract test suite (ATS)	R5	Yes
TS	34.229-5	Internet Protocol (IP) multimedia call control protocol based on Session Initiation Protocol (SIP) and Session Description Protocol (SDP); User Equipment (UE) conformance specification; Part 5: Protocol conformance specification using 5G System (5GS)	R5	Yes
TS	36.579-1	Mission Critical (MC) services over LTE; Part 1: Common test environment	R5	Yes

Type	Number	Title	Group	For publication?
TS	36.579-2	Mission Critical (MC) services over LTE; Part 2: Mission Critical Push To Talk (MCPTT) User Equipment (UE) Protocol conformance specification	R5	Yes
TS	36.579-4	Mission Critical (MC) services over LTE; Part 4: Test Applicability and Implementation Conformance Statement (ICS) proforma specification	R5	Yes
TS	36.579-5	Mission Critical (MC) services over LTE; Part 5: Abstract test suite (ATS)	R5	Yes
TS	36.579-6	Mission Critical (MC) services over LTE; Part 6: Mission Critical Video (MCVideo) User Equipment (UE) Protocol conformance specification	R5	Yes
TS	36.579-7	Mission Critical (MC) services over LTE; Part 7: Mission Critical Data (MCData) User Equipment (UE) Protocol conformance specification	R5	Yes

Annex A (informative): void

Annex B (informative): void

Annex C (informative): Change history

Change history							
Date	**TSG #**	**TSG Doc.**	**CR**	**Rev**	**Subject/Comment**	**Old**	**New**
2008-05					First draft based on SA1 request to SP-38, refined at SP-39. Spec list based on Specification Manager's whim.		0.0.0
2008-12					Update based on continuing evolution of Release 8 specs	0.0.0	0.1.0
2008-12	SP-42	SP-080724			Incorporation of comments from TSG CT: removal of §5.1 (substituted by two examples in annex A); indication in list of specs of those which do not progress beyond NGN R2 applicability.	0.1.0	0.2.0
2009-01					Update to specs list table resulting from further feedback; Correction to URL of TISPAN spec mapping following revamp of 3GPP web site	0.2.0	0.3.0
2009-03	CP-43	CP-090103			Presentationfor information	0.3.0	1.0.0
2009-03	SP-43	SP-090035			Presentation to TSG for approval	1.0.0	1.1.0
2009-03	SP-43				Approved	1.1.0	8.0.0
2009-05	SP-44	SP-090463	2	2	Annex mapping 3GPP2 specs to 3GPP common IMS specs	8.0.0	
		SP-090464	3		Revise scope		8.1.0
2009-06	SP-46	SP-090695	4	1	Correction to list of specifications	8.1.0	8.2.0
2010-03	SP-47	SP-100012	5		Update list of specs	8.2.0	8.3.0
			6		Update list of specs	8.3.0	
			7		Removal of information only relevant for earlier Release		9.0.0
2010-12	SP-50	SP-100871	8	1	Clarification of scope to indicate use of complete set of TSs/TRs is not obligatory in all IMS implementations	9.0.0	9.1.0
2011-03	SP-51				Automatic upgrade on Release freeze	9.1.0	10.0.0
2011-06	SP-52	SP-110298	11		Correction to list of specifications	10.0.0	10.1.0
2012-03	SP-55	SP-120114	14	1	Changes to list of Specs: common IMS	10.1.0	10.2.0
2013-03	SP-59	SP-130155	15		Update to list of specs	10.2.0	10.3.0
2013-03	SP-59	SP-130155	16		Update to list of specs	10.3.0	11.0.0
2014-06					Minor editorial	11.0.0	11.0.1

Change history							
Date	**Meeting**	**TDoc**	**CR**	**Rev**	**Cat**	**Subject/Comment**	**New version**
2016-03	SA#71	SP-160127	0017		F	Raise spec to Rel-12	12.0.0
2016-12	SA#74	SP-160975	0018	1	F	Update to list of specs	13.0.0
2017-03	SA#75					Promotion to Release 14 without technical change	14.0.0
2017-06	SA#76	SP-170461	0019		F	Replace list of specs with pointer to web page showing up to date list	14.1.0
2018-06	SA#80					Upgrade unchanged on freeze of Rel-15	15.0.0
2020-07	SA#88-e					Upgrade on freeze of Rel-16	16.0.0
2025-12	SA#110	SP-251609	0028	1	A	Correction of the IMS specifications list	16.1.0

History

Version	Date	Status
V16.0.0	August 2020	Publication
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